Distribuce

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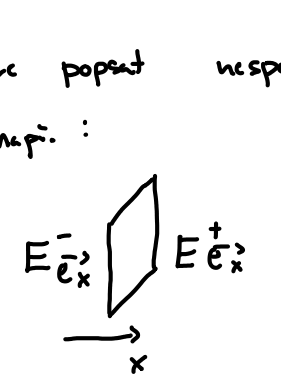
Distribuce v $\mathbb{R}^1$

→ chceme popsat singulární zdroje

- bodové, křivkové, plošné

→ chceme popsat nespojitost polí

- např.:



$\operatorname{div}\vec{E} = \frac{dE_x}{x}$

→ kvantová mechanika → vlastní stavy operátoru polohy $\hat{x}$

Problém: hustota bodového zdroje ρ v $x=0$

$$\Rightarrow \rho = \begin{cases} 0 & x \neq 0 \\ \text{něco} & x = 0 \end{cases}$$

a zároveň $\int_{\mathbb{R}} \rho \, dx = q$

→ problém s Lebesgueovským integrálem, pokud ρ je funkce

$\Rightarrow \int_{\mathbb{R}} \delta \, dx = 1$ → funkcionální definice

→ Schwartz 1950 – Fields medal

Spojitý lineární funkcionál na vhodném prostoru testovacích funkcí

$\mathcal{D}$ ... prostor testovacích funkcí

$\mathcal{D}'$ ... prostor distribucí

$\phi \in \mathcal{D}$, $T \in \mathcal{D}'$ $\Rightarrow$ $T : \mathcal{D} \to \mathbb{R}$

$$T(\phi) = \langle T, \phi \rangle \in \mathbb{R}$$

- spojitost $\phi_n \xrightarrow{\mathcal{D}} \phi \Rightarrow T(\phi_n) \xrightarrow{\mathbb{R}} T(\phi)$
- linearita $T(\alpha\phi_1 + \beta\phi_2) = \alpha T(\phi_1) + \beta T(\phi_2)$

$$T(\phi) = T_x \phi^x = \int_{\mathbb{R}} T_x \phi^x \, dx$$

$$\phi^x = \phi(x)$$

Prostory testovacích funkcí

Hladké funkce s kompaktním nosičem

$$\mathcal{D} = \{\phi \in C^\infty \mid \operatorname{supp}\phi \text{ kompaktní}\}$$

$\phi_n \xrightarrow{\mathcal{D}} \phi \Rightarrow \exists K$ (kompakt): $\forall n$

$\operatorname{supp}\phi_n \subset K$: $\phi_n^{[k]} \overset{K}{\rightrightarrows} \phi^{[k]}$ $\forall k \in \mathbb{N} \cup \{0\}$

$\mathcal{D}'$ ... distribuce s neomezeným nosičem

Hladké funkce rychle klesající

$$\mathcal{S} = \{\phi \in C^\infty \mid \forall m \ |x|^m \phi \text{ je omezená}\}$$

$\phi_n \xrightarrow{\mathcal{S}} \phi \Rightarrow \forall m,k$ $|x|^m \phi_n^{[k]} \overset{\mathbb{R}}{\rightrightarrows} |x|^m \phi^{[k]}$

$\mathcal{S}'$ ... Schwartzovské (temperované) distribuce

→ pomalu rostoucí

Hladké funkce

$$\mathcal{E} := \{\phi \in C^\infty(\mathbb{R})\}$$

$\phi_n \xrightarrow{\mathcal{E}} \phi \Rightarrow \forall K$ (kompakt) $\forall k$ $\phi_n^{[k]} \overset{K}{\rightrightarrows} \phi^{[k]}$

$\mathcal{E}'$ ... distribuce s kompaktním nosičem

Vztahy:

$\mathcal{D} \subset \mathcal{S} \subset \mathcal{E}$ $\qquad$ $\mathcal{E}' \subset \mathcal{S}' \subset \mathcal{D}'$

$L^2 = L^{2'}$ hodně, to ti říká Rieszova věta

Vlastnosti distribucí

Rovnost

$$T_1 \overset{\mathcal{D}'}{=} T_2 \iff \langle T_1, \phi \rangle = \langle T_2, \phi \rangle \quad \forall \phi \in \mathcal{D}$$

→ rovnost na množině $\Omega \subset \mathbb{R}$

$$T_1 \overset{\Omega}{=} T_2 \iff \langle T_1, \phi \rangle = \langle T_2, \phi \rangle \quad \forall \phi \in \mathcal{D}: \operatorname{supp}\phi \subset \Omega$$

Nosič distribuce

$$\operatorname{supp} T = \mathbb{R} \setminus \bigcup_{\Omega : T \overset{\Omega}{=} 0} \Omega$$

← identicky nulová distribuce

Konvergence

$$T_n \xrightarrow{\mathcal{D}'} T \iff \langle T_n, \phi \rangle \xrightarrow{\mathbb{R}} \langle T, \phi \rangle \quad \forall \phi \in \mathcal{D}$$

Příklady distribucí

Diracova δ funkce (distribuce)

$$\langle \delta, \phi \rangle = \phi(0)$$

ve fyzice: $\int_{\mathbb{R}} \delta(x)\phi(x)\,dx = \phi(0)$

Posunutá δ funkce

$$\langle \delta_a, \phi \rangle = \langle \delta(x-a), \phi(x) \rangle = \phi(a)$$

Regulární distribuce

- generování $L^1_{loc}(\mathbb{R})$ – lokálně integrabilní funkce na $\mathbb{R}$

$f \in L^1_{loc}(\mathbb{R})$

$$\langle T_f, \phi \rangle \text{“}=\text{”} \langle f, \phi \rangle = \int_{\mathbb{R}} f(x)\phi(x)\,dx$$

$$= \int_{\operatorname{supp}\phi} f(x)\phi(x)\,dx$$

je kompakt ⇒ existuje vždy

Heavisideova distribuce

$$\langle T_\theta, \phi \rangle = \langle \theta, \phi \rangle = \int_0^\infty \phi(x)\,dx$$

Derivace distribuce

Motivace

$$T'_f = T_{f'}$$

$$\langle T_{f'}, \phi \rangle = \int_{\mathbb{R}} f'\phi\,dx = \underbrace{[f\phi]_{-\infty}^{\infty}}_{0} - \int_{\mathbb{R}} f\phi'\,dx$$

supp ϕ je kompakt

$$= -\langle T_f, \phi' \rangle$$

Definice

$$\langle T', \phi \rangle = -\langle T, \phi' \rangle$$

$$\langle T'', \phi \rangle = \langle T, \phi'' \rangle$$

$$\vdots$$

$$\langle T^{(n)}, \phi \rangle = (-1)^n \langle T, \phi^{(n)} \rangle$$

Př

θ' ? $\Rightarrow$ $\langle \theta', \phi \rangle = -\langle \theta, \phi' \rangle$

$$= -\int_0^\infty \phi'(x)\,dx$$

$$= -[\phi]_0^\infty dx = \phi(0)$$

$$= \langle \delta, \phi \rangle$$

$\Rightarrow \theta' = \delta$

Př – Fyz

$$\operatorname{div}\vec{E} = \frac{\rho}{\varepsilon_0} \xrightarrow{1D} \frac{\partial E}{\partial x} = \frac{\rho}{\varepsilon_0}$$

Př

$$\langle \delta^{(n)}, \phi \rangle = (-1)^n \langle \delta, \phi^{(n)} \rangle = (-1)^n \phi^{(n)}(0)$$

Násobení distribucí hladkou funkcí

$T \in \mathcal{D}'$ $\quad f \in C^\infty$

$$\langle fT, \phi \rangle = \langle T, f\phi \rangle$$

Př

$$\langle f\delta, \phi \rangle = \langle \delta, f\phi \rangle = f(0)\phi(0) = f(0)\langle \delta, \phi \rangle$$

$$\langle f\delta', \phi \rangle = \langle \delta', f\phi \rangle = -\langle \delta, f'\phi + f\phi' \rangle =$$

$$= -f'(0)\phi(0) - f(0)\phi'(0) =$$

$$= -f'(0)\langle \delta, \phi \rangle + f(0)\langle \delta', \phi \rangle$$

$$f\delta' \overset{\mathcal{D}'}{=} f(0)\delta - f'(0)\delta$$

Př – Derivace nehladké funkce v $x=0$

$$f(x) = f_+(x)\theta(x) + f_-(x)\theta(-x)$$

$$\langle (f_+(x)\theta(x))', \phi \rangle = -\langle f_+(x)\theta(x), \phi' \rangle$$

$$= -\langle \theta(x), f_+(x)\phi' \rangle = -\int_0^\infty f_+(x)\phi'(x)\,dx$$

$$= -[f_+(x)\phi(x)]_0^\infty + \int_0^\infty f'_+(x)\phi(x)\,dx$$

$$= \langle f_+(0)\delta, \phi \rangle + \langle f'_+\theta, \phi \rangle$$

$$(f_+\theta)' \overset{\mathcal{D}'}{=} f_+(0)\delta + f'_+\theta$$

$$(f_-\theta(-x))' \overset{\mathcal{D}'}{=} -f_-(0)\delta + f'_-\theta(-x)$$

$$[f] = f_+(0) - f_-(0)$$

$f^{[1]} = f'_+\theta(x) + f'_-\theta(-x)$ regulární část derivace

$$f' = [f]\delta + f^{[1]}$$

$$f'' = [f]\delta' + [f']\delta + f^{[11]}$$

Př

$$xT \overset{\mathcal{D}'}{=} 0$$

$$\Updownarrow$$

$$T = c\delta \quad c \in \mathbb{R}$$

Dk "$\Downarrow$" vezměme $\omega \in \mathcal{D}$: $\omega(0) = 1$ je pevné

$\forall \phi \in \mathcal{D}$ $\exists \psi \in \mathcal{D}$: $\phi = \phi(0)\omega + x\psi$

$$\psi(0) = \lim_{x\to 0} \frac{\phi(x) - \phi(0)\omega(x)}{x}$$

$$\langle T, \phi \rangle = \phi(0)\langle T, \omega \rangle + \langle T, x\psi \rangle$$

$$= c\phi(0) + \underbrace{\langle xT, \psi \rangle}_{0} = c\langle \delta, \phi \rangle$$

$$\Rightarrow T \overset{\mathcal{D}'}{=} c\delta$$

Př – distribuce $\ln|x|$

Platí $\ln|x| \in L^1_{loc}(\mathbb{R})$?

$$\lim_{\varepsilon\to 0} \int_\varepsilon^1 \ln|x|\,dx = \lim_{\varepsilon\to 0} [x\ln x - x]_\varepsilon^1$$

$$= \lim_{\varepsilon\to 0} (-\varepsilon\ln\varepsilon - 1 + \varepsilon) = -1$$

$$\Rightarrow \ln|x| \in L^1_{loc}(\mathbb{R})$$

$$\langle \ln|x|, \phi \rangle = \int_{\mathbb{R}} \ln|x|\phi(x)\,dx$$

Př – distribuce v.p. $\frac{1}{x}$

v.p. ... valeur principal – hlavní hodnota

$$\left\langle \text{v.p.}\frac{1}{x}, \phi \right\rangle = \lim_{\varepsilon\to 0} \int_{\mathbb{R}\setminus(-\varepsilon,\varepsilon)} \frac{\phi(x)}{x}\,dx$$

$$= \lim_{\varepsilon\to 0} \int_\varepsilon^\infty \frac{\phi(x) - \phi(-x)}{x}\,dx$$

$$\langle (\ln|x|)', \phi \rangle = -\langle \ln|x|, \phi' \rangle =$$

$$= -\int_{\mathbb{R}} \ln|x|\phi'(x)\,dx = -\lim_{\varepsilon\to 0}\left(\int_{-\infty}^{-\varepsilon} \dots + \int_\varepsilon^\infty \dots\right)$$

$$\lim_{\varepsilon\to 0} \int_\varepsilon^\infty \ln|x|\phi(x)\,dx = \lim_{\varepsilon\to 0}\left([\ln|x|\,\phi(x)]_\varepsilon^\infty - \int_\varepsilon^\infty \frac{\phi(x)}{x}\,dx\right)$$

$$= \lim_{\varepsilon\to 0}\left(-\ln|\varepsilon|\phi(\varepsilon) - \int_\varepsilon^\infty \frac{\phi(x)}{x}\,dx\right)$$

$$\lim_{\varepsilon\to 0} \int_{-\infty}^{-\varepsilon} \ln|x|\phi'(x)\,dx = \lim_{\varepsilon\to 0}\left(\ln|\varepsilon|\phi(-\varepsilon) - \int_\infty^{-\varepsilon} \frac{\phi(x)}{x}\,dx\right)$$

$$\underset{\substack{\Downarrow \\ 0}}{-\ln|\varepsilon|\,\varepsilon}\ \frac{\phi(\varepsilon) - \phi(-\varepsilon)}{\varepsilon}$$

$$\Rightarrow (\ln|x|)' \overset{\mathcal{D}'}{=} \text{v.p.}\frac{1}{x}$$

Substituce v distribuci

Motivace skrze regulární distribuce

$T_f \to T_{f\circ g}$, tj. $T_{f(y)} \to T_{f(g(x))}$

$$\langle T_{f(g(x))}, \phi(x) \rangle = \int_{\mathbb{R}} f(g(x))\phi(x)\,dx$$

$$\left[g(x) = y \Rightarrow dx = \frac{1}{g'(g^{-1}(y))}dy\right]$$

g je prostá

$$= \int_{\mathbb{R}} f(y)\phi(g^{-1}(y))\frac{dy}{|g'(g^{-1}(y))|}$$

$\Rightarrow$ pro $T \in \mathcal{D}'$

$$\langle T\circ g, \phi \rangle = \langle T(f), \phi \rangle = \left\langle T, \frac{\phi\circ g^{-1}}{|g'\circ g^{-1}|} \right\rangle$$

Př

$$\langle \delta\circ g, \phi \rangle = \langle \delta(g), \phi \rangle = \left\langle \delta, \frac{\phi\circ g^{-1}}{|g'\circ g^{-1}|} \right\rangle$$

$$= \frac{\phi\circ g^{-1}(0)}{|g'\circ g^{-1}(0)|} = \sum_{x_i : g(x_i) = 0} \frac{\phi(x_i)}{|g'(x_i)|}$$

$$= \sum_{x_i : g(x_i) = 0} \frac{1}{|g'(x_i)|}\langle \delta_{x_i}, \phi \rangle$$

$$\delta\circ g = \delta(g(x)) = \sum_{x_i} \frac{1}{|g'(x_i)|}\underset{\substack{\uparrow \\ \delta(x-x_i)}}{\delta_{x_i}}$$

Greenovy funkce v 1D

→ řešení pro bodový zdroj

⇒ superp. → řešení pro lib. rozložení

$$L_x f(x) = g(x)$$

$$L_x G = \delta$$

$$L_x G(x,x') \overset{\mathcal{D}'}{=} \delta(x,x')$$

$$f(x) = \langle G(x,x'), g(x') \rangle$$

$$L_x f(x) = \langle L_x G(x,x'), g(x') \rangle = \langle \delta(x-x'), g(x') \rangle = g(x)$$

$$f(x) = \int G(x,x')g(x')\,dx'$$

$$L_x f = \int L_x G(x,x')g(x')\,dx' = g(x)$$

- pozn. $\underset{\text{jádro}}{L_x h(x) = 0}$

$$G(x,x') \to \tilde{G}(x,x') = G(x,x') + h(x)$$

→ nejednoznačnosti se zbavíme o.p.

- porovnání

$$L_x = \frac{d}{dx} \Rightarrow G(x,x') = \theta(x-x')$$

$$L_x\theta(x-x') = \delta(x-x')$$

Problém:

L_x je lineární samosdružený dif. operátor

$(Af, g) = (f, A^\dagger g)$ $\qquad$ $A = A^\dagger$

$I \subset \mathbb{R}$ ... interval řešení $\qquad$ $\frac{d}{dx} \to i\frac{d}{dx}$

$B_{\partial I}$ ... o.p

$B_{\partial I} f = C \Rightarrow a$

- $f|_{\partial I} = 0$ ... Dirichletovy
- $f'|_{\partial I} = 0$ ... Neumannovy

Věta

Nechť $L_x = \sum_{n=0}^{k} a_n(x)\frac{d^n}{dx^n}$ lin. samosdružený dif. op. ř. k,

$I \subset \mathbb{R}$, $B_{\partial I}$.

Pak nechť

1) $\exists g_y^\pm(x)$ takové, že:

$$L_x g_y^+(x) = 0 \quad x \geq y$$

$$L_x g_y^-(x) = 0 \quad x \leq y$$

2) $B_{\partial I+} g^+ = 0$ $\quad$ $B_{\partial I-} g^- = 0$ $\quad$ $\partial I^\pm$ pravá/levá hranice

3) $\forall n \in \{0, \dots, k-2\}$ $\quad g_y^{+(n)}(y) = g_y^{-(n)}(y)$

4) $g_y^{+(k-1)}(y) - g_y^{-(k-1)}(y) = \frac{1}{a_k(y)}$

$$\Rightarrow G(x,y) = g_y^+(x)\theta(x-y) + g_y^-(x)\theta(y-x)$$

Dk

$$\frac{d}{dx}G(x,y) = g_y'^+(x)\theta(x-y) + g_y'^-(x)\theta(y-x)$$

$$+ g_y^+(x)\delta(x-y) + g_y^-(x)\delta(y-x)$$

$$= g_y^+(x)\delta(x-y) - g_y^-(x)\delta(x-y)$$

$$\frac{d^k}{dx^k}G(x,y) = g_y^{+(k)}(x)\theta(x-y) + g_y^{-(k)}(x)\theta(y-x)$$

$$+ \text{“}g^{(k-1)}\delta + \cancel{g^{(k-2)}\delta'} + \dots$$

$$+ \cancel{g\,\delta^{(k-1)}}\text{”}$$

$$\left(g_y^{+(k-1)}(x) - g_y^{-(k-1)}(x)\right)\delta(x-y)$$

$$= \frac{1}{a_k(y)}\delta(x-y)$$

$$L_x f(x) = g(x) \qquad f(x) = \int G(x,y)g(y)\,dy$$

Př:

$L_x = -\frac{d^2}{dx^2} + k^2$ $\quad$ $I = (0,d)$ $\quad$ $B_{\partial I} f = f|_{\partial I} = 0$

$L_x g^\pm = 0 \Rightarrow$ FS: e^{-kx}, e^{kx}

$$g_y^\pm(x) = A^\pm(y)e^{kx} + B^\pm(y)e^{-kx}$$

$B_{\partial I}$: $\quad A^+(y)e^{kd} + B^+(y)e^{-kd} = 0$

$$\Rightarrow B^+(y) = -A^+(y)e^{2kd}$$

$$A^-(y) = -B^-(y)$$

spoj $g^\pm$:

$$A^+(y)(e^{ky} - e^{2kd}e^{-ky}) = A^-(y)(e^{ky} - e^{-ky})$$

skok $g^{\pm\prime}$:

$$kA^+(y)(e^{ky} + e^{2kd}e^{-ky})$$

$$- kA^-(y)(e^{ky} + e^{-ky}) = \frac{1}{a_2(y)} = -1$$

$$\Rightarrow A^+(y) = -\frac{e^{-kd}}{2k}\frac{\sinh(ky)}{\sinh(kd)}$$

$$A^-(y) = \frac{1}{2k}\frac{\sinh(k(y-d))}{\sinh(kd)}$$

$$G(x,y) = -\frac{1}{k}\frac{\sinh(ky)}{\sinh(kd)}\sinh(k(x-d))\,\theta(x-y)$$

$$- \frac{1}{k}\frac{\sinh[k(y-d)]}{\sinh(kd)}\sinh(kx)\,\theta(y-x)$$

$$G_0(x,y) = \lim_{k\to 0} G(x,y) = -\frac{yx}{d} + y\theta(x-y) + x\theta(y-x)$$

často se setkáme s $L_x = -\frac{d^2}{dx^2} + \frac{1}{x}$ na $(0,d)$

⇒ řešíme na (ε, d) a na konci provedeme limitu $\varepsilon \to 0^+$ a ověříme omezení.

L_x ... samosdružený lineární diferenciální operátor

$$L_x\psi_n = \lambda_n\psi_n, \quad I \subset \mathbb{R}, \quad B_{\partial I}$$

Relace ortogonality:

$$\int_I \psi_n(x)\bar{\psi}_m(x)\,dx = \delta_{nm}$$

Relace úplnosti

$$\sum_{n=0}^{\infty}\psi_n(x)\bar{\psi}_n(x') = \delta(x-x')$$

$$\to \int \sum_{n=0}^{\infty}\psi_n(x)\underbrace{\bar{\psi}_n(x)f(x)\,dx}_{c_n} = \int \delta(x-x')f(x)\,dx$$

$$= f(x')\,dx$$

$\Rightarrow$ projektor $\sum |\psi_n\rangle\langle\psi_n|$

$\Rightarrow$ Greenova funkce

$$G(x,y) = \sum_{n=0}^{\infty} c_n(y)\psi_n(x)$$

$$L_x G(x,y) = \sum_{n=0}^{\infty} c_n(y)\lambda_n\psi_n(x) \overset{?}{=} \delta(x-y) \quad \Big|\bar{\psi}_m(x)\Big|\int_I dx$$

$$\downarrow$$

$$\int \sum_{n=0}^{\infty} c_n(y)\lambda_n\bar{\psi}_m\psi_n\,dx = \int \bar{\psi}_m(x)\delta(x-y)\,dx$$

$$\sum_{n=0}^{\infty} c_n(y)\lambda_n\delta_{mn} = \bar{\psi}_m(y)$$

$$\lambda_m c_m(y) = \bar{\psi}_m(y)$$

$$\Rightarrow G(x,y) = \sum_{n=0}^{\infty}\frac{\psi_n(x)\bar{\psi}_n(y)}{\lambda_n}$$

$= 0$?

Co když existuje homo řešení?

$L_x = -\frac{d^2}{dx^2}$ $\quad$ $I = S^1$ $\quad$ $x \in (0,1)$, $\psi(0) = \psi(1)$

$L_x c = 0$ → jakákoli konstanta řeší tento problém

$\Rightarrow \lambda_0 = 0$ $\quad$ ⇒ zahodíme nultý člen:

$$\tilde{G}(x,y) = \sum_{n=1}^{\infty}\frac{\psi_n(x)\bar{\psi}_n(x)}{\lambda_n}$$

$$\Rightarrow L_x\tilde{G}(x,y) = \delta(x-y) - \underbrace{\psi_0(x)\bar{\psi}_0(y)}_{\text{nějaká konstanta}}$$

musíme zúplnit